SCIENCE AND TECHNOLOGY

STANDARD TEN

PART - 1

The Constitution of India

Chapter IV A

Fundamental Duties

ARTICLE 51A

Fundamental Duties- It shall be the duty of every citizen of India–

(a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;

(b) to cherish and follow the noble ideals which inspired our national struggle for freedom;

(c) to uphold and protect the sovereignty, unity and integrity of India;

(d) to defend the country and render national service when called upon to do so;

(e) to promote harmony and the spirit of common brotherhood amongst all the people of India transcending religious, linguistic and regional or sectional diversities, to renounce practices derogatory to the dignity of women;

(f) to value and preserve the rich heritage of our composite culture;

(g) to protect and improve the natural environment including forests, lakes, rivers and wild life and to have compassion for living creatures;

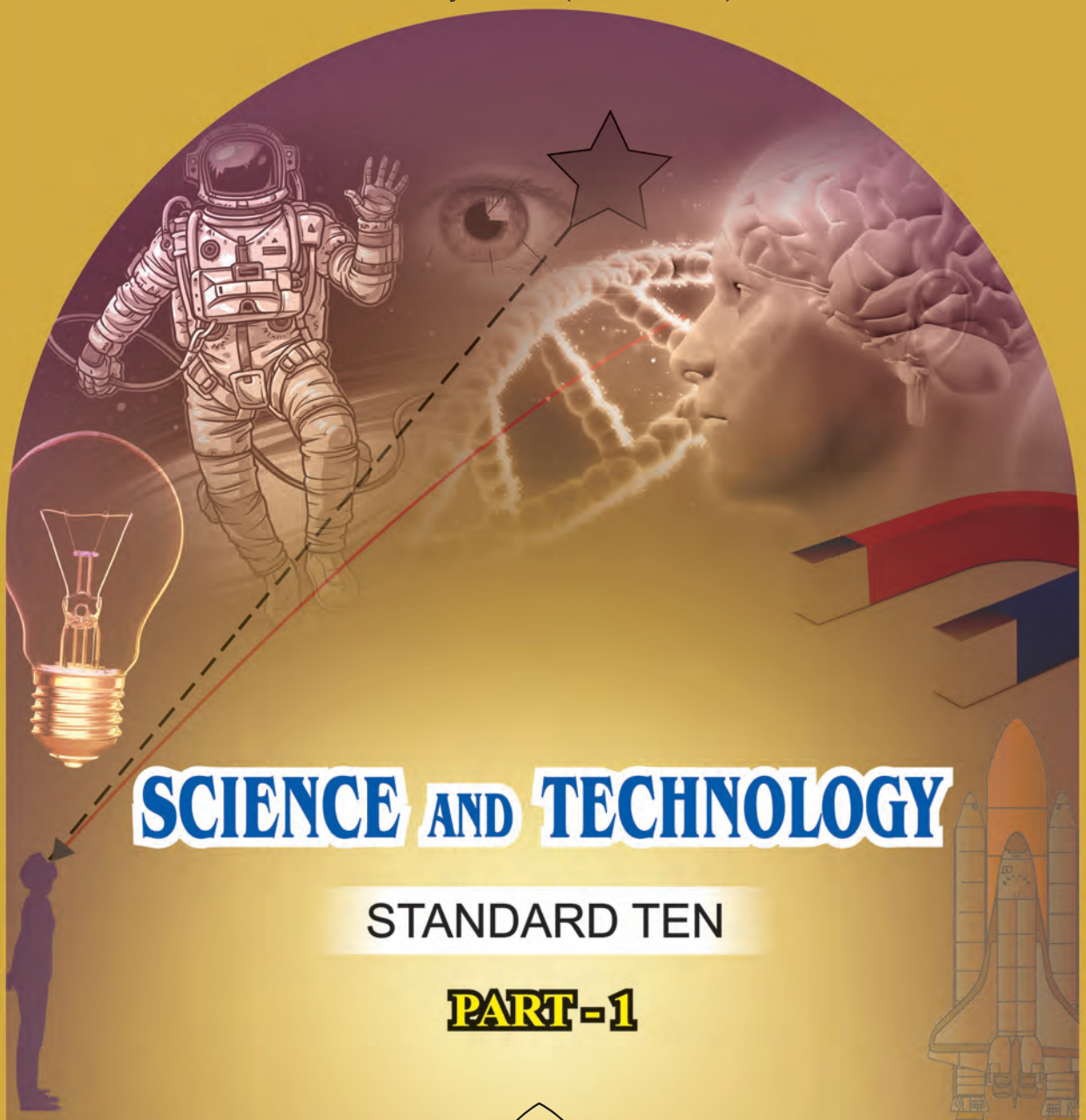
(h) to develop the scientific temper, humanism and the spirit of inquiry and reform;

(i) to safeguard public property and to abjure violence;

(j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement;

(k) who is a parent or guardian to provide opportunities for education to his child or, as the case may be, ward between the age of six and fourteen years.

Permission is granted for enforcing this textbook from the academic year 2018-19 in the meeting, held on the date 29.12.2017, of the coordination committee constituted by the Government resolution No: Abhyas-2116/(Pra.kra.43/16) S.D-4 dated 25.4.2016



SCIENCE AND TECHNOLOGY

STANDARD TEN

PART - 1

Maharashtra State Bureau of Textbook Production and Curriculum Research, Pune.

HMTQBV

The digital textbook can be obtained through DIKSHA App on your smartphone by using the Q.R.Code given on title page of the textbook and useful audio-visual teaching-learning material of the relevant lesson will be available through the Q.R. Code given in each lesson of this textbook.

First Edition : 2018
Reprint : 2022

© Maharashtra State Bureau of Textbook Production and Curriculum Research, Pune - 411 004.

The Maharashtra State Bureau of Textbook Production and Curriculum Research reserves all rights relating to the book. No part of this book should be reproduced without the written permission of the Director, Maharashtra State Bureau of Textbook Production and Curriculum Research, 'Balbharati', Senapati Bapat Marg, Pune 411004.

Science Subject Committee :

Dr. Chandrashekhar Murumkar, Chairman
Dr. Dilip Sadashiv Joag, Member
Dr. Sushama Dilip Joag, Member
Dr. Pushpa Khare, Member
Dr. Imtiaz Sirajuddin Mulla, Member
Dr. Jaydeep Vinayak Sali, Member
Dr. Abhay Jere, Member
Dr. Sulabha Nitin Vidhate, Member
Smt. Mrinalini Desai, Member
Shri. Gajanan Suryawanshi, Member
Shri. Sudhir Yadavrao Kamble, Member
Smt. Dipali Dhananjay Bhale, Member
Shri. Rajiv Arun Patole, Member-Secretary

Science Study Group :

Dr. Prabhakar Nagnath Kshirsagar
Dr. Vishnu Vaze
Dr. Prachi Rahul Choudhary
Dr. Shaikh Mohammed Waquioddin H.
Dr. Ajay Digambar Mahajan
Dr. Gayatri Gorakhnath Choukade
Shri. Prashant Panditrao Kolse
Shri.Sandip Popatlal Chordiya
Shri. Sachin Ashok Bartakke
Smt. Shweta Dilip Thakur
Shri. Rupesh Dinkar Thakur
Shri. Dayashankar Vishnu Vaidya
Shri. Sukumar Shrenik Navale
Shri. Gajanan Nagoraoji Mankar
Shri. Mohammed Atique Abdul Shaikh
Smt. Anjali Lakshmikant Khadke
Smt. Manisha Rajendra Dahivelkar
Smt. Jyoti Medpilwar
Smt. Dipti Chandansingh Bisht
Smt. Pushpalata Gawande
Smt. Anita Patil
Smt. Kanchan Rajendra Sorate
Shri. Rajesh Vamanrao Roman
Shri. Nagesh Bhimsevak Telgote
Shri. Shankar Bhikan Rajput
Shri. Manoj Rahangdale
Shri. Hemant Achyut Lagvankar
Smt. Jyoti Damodar Karane
Shri. Vishwas Bhave

Cover and Illustrations :
Shri. Vivekanand Shivshankar Patil
Ashana Advani

Typesetting :
DTP Section, Textbook Bureau, Pune

Coordination :
Rajiv Arun Patole
Special Officer for Science

Translation :
Dr. Sushma Dilip Joag
Dr. Pushpa Khare
Dr. Dilip Sadashiv Joag

Scrutiny :
Dr. Imtiaz S. Mulla
Dr. Jaydeep Vinayak Sali

Paper :
70 GSM Creamwove

Print Order :

Printer :

Production :
Sachchitanand Aphale
Chief Production Officer
Rajendra Vispute
Production Officer

Publisher :
Vivek Uttam Gosavi
Controller
Maharashtra State Textbook Bureau,
Prabhadevi, Mumbai - 400 025.

The Constitution of India

Preamble

WE, THE PEOPLE OF INDIA, having solemnly resolved to constitute India into a SOVEREIGN SOCIALIST SECULAR DEMOCRATIC REPUBLIC and to secure to all its citizens:

JUSTICE, social, economic and political;

LIBERTY of thought, expression, belief, faith and worship;

EQUALITY of status and of opportunity; and to promote among them all

FRATERNITY assuring the dignity of the individual and the unity and integrity of the Nation;

IN OUR CONSTITUENT ASSEMBLY this twenty-sixth day of November, 1949, do HEREBY ADOPT, ENACT AND GIVE TO OURSELVES THIS CONSTITUTION.

NATIONAL ANTHEM

Jana-gana-mana-adhināyaka jaya hē
Bhārata-bhāgya-vidhātā,

Panjāba-Sindhu-Gujarāta-Marāthā
Drāvida-Utkala-Banga

Vindhya-Himāchala-Yamunā-Gangā
uchchala-jaladhi-taranga

Tava subha nāmē jāgē, tava subha āsisa māgē,
gāhē tava jaya-gāthā,

Jana-gana-mangala-dāyaka jaya hē
Bhārata-bhāgya-vidhātā,

Jaya hē, Jaya hē, Jaya hē,
Jaya jaya jaya, jaya hē.

PLEDGE

India is my country. All Indians are my brothers and sisters.

I love my country, and I am proud of its rich and varied heritage. I shall always strive to be worthy of it.

I shall give my parents, teachers and all elders respect, and treat everyone with courtesy.

To my country and my people, I pledge my devotion. In their well-being and prosperity alone lies my happiness.

Preface

Dear students

Welcome to Std X. We have great pleasure in offering you this Science and Technology textbook based on the new syllabus. From the primary level till today, you have studied science from various textbooks. In this textbook, you will be able to study the fundamental concepts of science and technology from a different point of view through the medium of the different branches of Science.

The basic purpose of this textbook Science and Technology Part-1 can be said to be 'Understand and explain to others' the science and technology that relates to our everyday life. While studying the concepts, principles and theories in science, do make the effort to understand their connection with day to day affairs. While studying from this textbook, use the sections 'Can you recall?' and 'Can you tell?' for revision. You will learn science through the many activities given under the titles such as 'Observe and discuss' and 'Try this' or 'Let's try this. Make sure that you perform all these activities. Activities like 'Use your brain power!', 'Research', 'Think about it' will stimulate your power of thinking.

Many experiments have been included in the textbook. Carry out these experiments yourself, following the given procedure and making your own observations. Ask your teachers, parents or classmates for help whenever you need it. Interesting information which reveals the science underlying the events we commonly observe, and the technology developed on its basis, has been given in details in this textbook through several activities. In this world of rapidly developing technology, you have already become familiar with computers and smartphones. While studying the textbook, make full and proper use of the devices of information communication technology, which will make your studies easier. For more effective studies, you can avail additional audio-visual material for each chapter using the Q.R code through an App.This will definitely help you in your studies.

While carrying out the given activities and experiments, take all precautions with regard to handling apparatus, chemicals, etc. and encourage others to take the same precautions.

It is expected that while carrying out activities or observation involving plants and animals, you will also make efforts towards conservation of the environment. You must of course take all the care to avoid causing any harm or injury to them.

Do tell us about the parts that you like, as well as about the difficulties that you face as you read and study and understand this textbook.

Our best wishes for your academic progress.

(Dr.Sunil Magar)

Director

Maharashtra State Bureau of Textbook and Curriculum Research, Pune

Pune

Date : 18 March 2018, Gudhipadva

Indian Solar Year : 27 Phalgun 1939

For Teachers

- In Standards I to V we have told the simple science in day to day life through the study of surroundings. In VI to VIII standard we have given brief introduction to science. In the textbook 'Science and Technology' for standard IX we have given the relation between science and technology.
- The real objective of science education is to learn to be able to think logically and with discretion about events that are happening around us.
- In view of the age group of Std X students, it would be appropriate, in the process of science education, to give freedom and scope to the students' own curiosity about the events of the world, their propensity to go looking for the causes behind them and to their own initiative and capacity to take the lead.
- As experimental skills are necessary for observation, logic, estimation, comparison and application of information obtained in science education, deliberate efforts must be made to develop these skills while dealing with laboratory experiments given in the textbook. All observations that the students have noted should be accepted, and then they should be helped to achieve the expected results.
- These two years in middle school lay the foundation of higher education in Science. Hence, it is our responsibility to enrich and enhance student's interest in science. You all will of course always actively pursue the objective of imbuing them with a scientific temper in them and developing their creativity and along with internet and skill.
- You can use 'Let's recall' to review the previous knowledge required for a lesson and 'Can you tell?' to introduce a topic by eliciting all the knowledge that the students already have about it from their own reading or experience. You may of course use any of your own activities or questions that occur to you for this purpose. Activities given under 'Try this' and 'Let's try this' help to explain the content of the lesson. The former are for students to do themselves and the latter are those that you are expected to demonstrate. 'Use your brain power!' is meant for application of previous knowledge for the new lesson, and 'Always remember' gives important suggestions/information or values. 'Research', 'Find out', 'Do you know?', 'Introduction to scientists' and 'Institutes at work' are meant to give some information about the world outside the textbook and to develop the habit of doing independent reference work to obtain additional information.
- This textbook is not only meant for reading and explaining in the classroom but is also for guiding students to learn the methods of gaining knowledge by carrying out the given activities. An informal atmosphere in the classroom is required to achieve the aims of this textbook. maximum number of students should be encouraged to participate in discussions, experiments and activities. Special efforts should be made to organise presentations or report-reading in the class based on students' activities and projects, besides observing of Science Day and other relevant occasions/ days.
- The science and technology content of the textbook has been complemented with Information Communication Technology. These activities are to be conducted under your guidance while learning various new scientific concepts.

Front and back covers : Pictures of various activities, experiments and concepts in the book.

DISCLAIMER Note : All attempts have been made to contact copy righters (©) but we have not heard from them. We will be pleased to acknowledge the copy right holder (s) in our next edition if we learn from them.

Competency Statements

The students are expected to achieve the following competency level after studying the text book Science and Technology Part 1

Motion, Force and Machines

* To be able to explain the scientific reasons behind various phenomenal on the basis of relationship between gravitational force and motion.
* To be able to write formulae describing the relations between gravitation and motion and using these solve various numerical problems.

Energy

* To adapt an environment friendly lifestyle taking into account the grave effects of energy crisis and to encourage others to adapt it.
* To prepare , use and repair the equipments based on energy.
* To verify the laws of current electricity and to draw conclusions based on them
* To develop to solve numerical problems based on effects of current electricity.
* To observe various apparatus based on effects of current electricity and explain their functions with reasons.
* To give a scientific explanation of the images formed by lenses by drawing accurate ray diagrams.
* To explain properties of light, the images formed by lenses and their use in different equipments used in day to day life.
* To find out the focal length of a lens using given data.
* To study defects of vision in human eye and their remedies
* To draw neat and labelled diagram of human eye.

Substances in our use

* To explain systematic classifications of elements and their positions in the periodic table.
* To identify type of chemical reaction in two components.
* To verify chemical reaction experimentally and draw conclusions.
* To correct the chemicals equation which is incomplete or wrong.
* To verify the properties of carbon compounds through experiments.
* To take proper care while performing the experiments and handling of the apparatus considering the effects of chemical reactions on human health.
* To guide the society through scientific attitude about the use of carbon compounds in daily life.
* To understand the relationship between chemical reaction of metals in daily life and use them to solve various problem.

The Universe

* To analyse the information obtained from space research and remove superstitions prevailing in society.
* To review the contribution made by India to space research.
* To search for future opportunities in the field of space research.

Information Communication technology (ICT)

* To use information communication Technology in day today life.
* To share the information about science and technology by using the internet.
* To explain amazing that have occurred fields by using information communication technology

Index

Sr No.	Title of Lesson	Page No.
1.	Gravitation	1
2.	Periodic Classification of Element	16
3.	Chemical reactions and equations	30
4.	Effects of electric current	47
5.	Heat	62
6.	Refraction of light	73
7.	Lenses	80
8.	Metallurgy	93
9.	Carbon compounds	110
10.	Space Missions	135

Academic Planning

Two separate books have been prepared for Science and technology. Science and technology part 1 contains ten chapters mainly related to physics and chemistry. While thinking about science and technology, it is expected that an integrated approach will be taken while teaching and a connection will be made between different components of science and technology. In previous standards, we have studied various topics in science and technology together. For technical case two separate books science and technology part 1 and part 2 have been prepared, but it is necessary that an integrated perspective be taken while teaching.

Out of the ten chapters included in text book science and technology part 1, the first five chapters are expected to be taught in the first session while the next five chapters in the second session. At the end of a session a written examination for 40 marks and a practical examination for ten marks should be conducted. Exercises and projects have been given at the end of every chapters in the text book.

In view of evaluation, representative questions similar to those in the activity sheets of language books are given in exercises. You may make similar other questions for your use. The students should be evaluated based on these questions detailed information above to this will be given in separate evaluation scheme.